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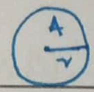
{ input, input → function → output } { input → function → output, not a function, output }



Area is function of r

Functions:

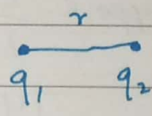
{ function is like a machine that Take inputs and give unique output }

Explaination:

Input → function → output

Defination:

- Function is relation b/w 2 or more quantities. Such that every value of 1st quantity (Independent Variable) is related to unique value of 2nd quantity (dependent variable)



$F = f(q_1, q_2, r)$

Home

distance (time, velocity)

Single variable function:

- A function with only 1 independent variable is called single variable function. example: Area of circle is single variable function of radius. (Area = f(radius))

Area → Dependent variable; radius → independent variable

price = f(stock)
price → dependent; stock → independent

boiling point = f(height)

$y = f(x)$

acceleration = f(force)

(not include

[include

Multivariable functions

- Multivarible functions are those function that contain 2 or more independent quantities

$F = f(q_1, q_2, r)$
F → dependents; q_1, q_2, r → independents

denote: $z = f(x, y)$

Denote: $A = f(x)$, $P = f(s)$

- $y = f(x) = x^2$
  $y = \sin x$
  $y = e^x$

y = f(x) (function)
y → dependent variable; x → independent variable

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Domain & Range of Function:

∵ by allowed we mean that
case 1 - no zero is denominator.
case 2 - no negative number in $\sqrt{-ve}$ $\sqrt[odd/even]{-}$

Domain of Function: {all real numbers that gives finite real number.}
Set of all allowed inputs (independent variables)

Range of Function:
Set of all possible outputs

Q1: Find domain of the function.

1 - $y = f_1(x) = \frac{1}{x}$

$Dom(f_1(x)) = \mathbb{R} / \{0\} = \mathbb{R} - \{0\}$

in interval form $= (-\infty, 0) \cup (0, \infty)$

Real line: $(-\infty, 0)$ $(0, \infty)$; $-\infty$ ... ∞

• All non-zero real number
$= \{x \in \mathbb{R} \mid x \neq 0\}$

5 - $f(x) = \frac{1}{2}x^2 + x$ $Dom\, f_5(x) =$ All $\mathbb{R}$ numbers

6 - $y = f_6(x) = \frac{\sin x}{\cos x} \Rightarrow Dom\, f_6(x) = \mathbb{R} / \{\pm(2n+1)\frac{\pi}{2}$ $n \in \mathbb{Z}$

$\cos x = 0$

$x = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}, \pm\frac{7\pi}{2} \ldots$ $x = (2x+1)\frac{\pi}{2}$ $n \in \mathbb{Z}$

2 - $y = f_2(x) = \frac{2}{x-2}$

$Dom(f_2(x)) = \mathbb{R} / \{2\}$

3. $y = f_3(x) = \frac{1}{x^2 - 1}$

$Dom\, f_3(x) = \mathbb{R} / \{\pm 1\}$ Real line $(-\infty, 1)$ $(-1, 1)$ $(1, \infty)$; $-\infty$ -1 0 1 ∞

Summary:
$y = f(x) = \frac{P(x)}{q(x)}$

$Dom\, f(x) = \{x \in \mathbb{R} \mid q(x) \neq 0\}$

• For domain suppose $q(x) = 0$ & solve for x.

4. $y = f_4(x) = \frac{x+1}{x^3 - x^2 + 5x}$

for domain

$x^3 - x^2 + 5x = 0$, $Dom\, f_4(x) = \mathbb{R} / \{0\}$

$x(x^2 - x + 5) = 0$

$x^2 - x + 5 = 0$, $x = 0$

using quadratic formula

$x = \frac{-(-1) \pm \sqrt{1 - 4(5)}}{2} \Rightarrow x = \frac{1 \pm \sqrt{-19}}{2}$ → complex number (not a part of real number

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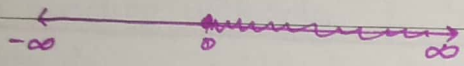
Question when underroot are involved.

For domain quantity inside the underroot should be greater than and equal to zero.

1) $y = f_1(x) = \sqrt{x}$

For domain $x \geq 0$

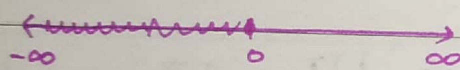
$Dom(f_1(x)) = \{x \in \mathbb{R} \mid x \geq 0\}$
$= [0, \infty)$



2) $y = f_2(x) = \sqrt{-x}$

For domain $-x \geq 0$ or $x \leq 0$

$Dom(f_2(x)) = \{x \in \mathbb{R} \mid x \leq 0\} = (-\infty, 0]$



3) $y = f_3(x) = \sqrt{x-1}$

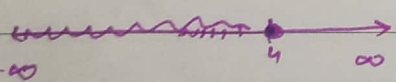
$x - 1 \geq 0 \Rightarrow x \geq 1$

$Dom\, f_3(x) = \{x \in \mathbb{R} \mid x \geq 1\}$
$= [1, \infty)$

4) $y = f_4(x) = \sqrt{4-x}$

$4 - x \geq 0 \Rightarrow -x \geq -4 \Rightarrow x \leq 4$

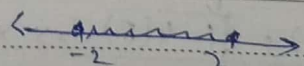
$Dom\, f_4(x) = \{x \in \mathbb{R} \mid x \leq 4\} = (-\infty, 4]$



5) $y = f_5(x) = \sqrt{4 - x^2}$

$4 - x^2 \underbrace{\geq 0}_{\text{non-negative}} \Rightarrow 4 \geq x^2 \Rightarrow \sqrt{x^2} \leq \sqrt{4} \Rightarrow |x| \leq 2 \Rightarrow -2 \leq x \leq 2$

$Dom\, f_5(x) = \{x \in \mathbb{R} \mid -2 \leq x \leq 2\} \Rightarrow [-2, 2]$



6) $f(x) = \sqrt{5 - x - x^2}$

For domain

$5 - x - x^2 \geq 0$

completing square

$\cancel{-(-5 + x + x^2) \geq 0}$

$x^2 + x - 5 \leq 0$

$x + x^2 \leq 5$

$x^2 + x + \left(\frac{1}{2}\right)^2 \leq 5 + \left(\frac{1}{2}\right)^2$

$\left(x + \frac{1}{2}\right)^2 \leq 5 + \frac{1}{4}$

$\left(x + \frac{1}{2}\right)^2 \leq \frac{21}{4}$

taking square root

$\sqrt{\left(x + \frac{1}{2}\right)^2} \leq \sqrt{\frac{21}{4}}$

$\left|x + \frac{1}{2}\right| \leq \frac{\sqrt{21}}{2}$

Suppose $y = x + \frac{1}{2}$

$|y| \leq \sqrt{21}/2$

$-\sqrt{21}/2 \leq y \leq \sqrt{21}/2$

$-\sqrt{21}/2 \leq x + \frac{1}{2} \leq \sqrt{21}/2$

$-\sqrt{21}/2 - \frac{1}{2} \leq x \leq \sqrt{21}/2 - \frac{1}{2}$

$\frac{-\sqrt{21} - 1}{2} \leq \; \leq \frac{\sqrt{21} - 1}{2}$

in general

$y = f(x) = \sqrt{g(x)}$

For domain

$g(x) \geq 0$

and solve this inequality for x

• Set of all value x on which function is defined → domain

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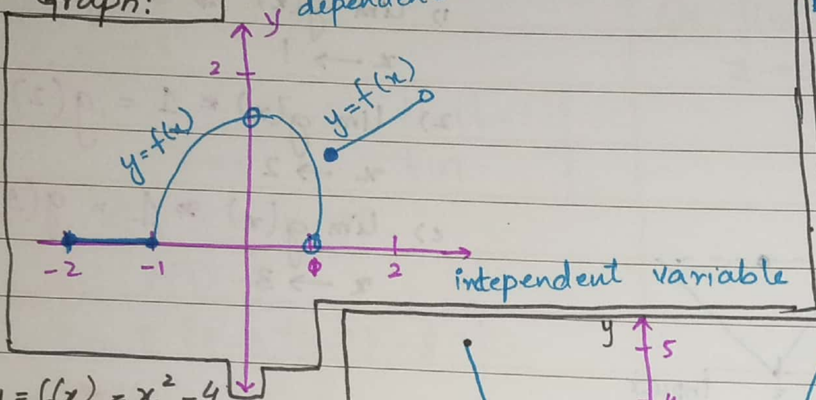
Graph:



Domain & range from graph

Dom $(f(x)) = [-2, 0) \cup (0, 2)$

Shadow of graph on x-axis → domain
light will pass from hole (bulb)

Range: Shadow of graph on y-axis

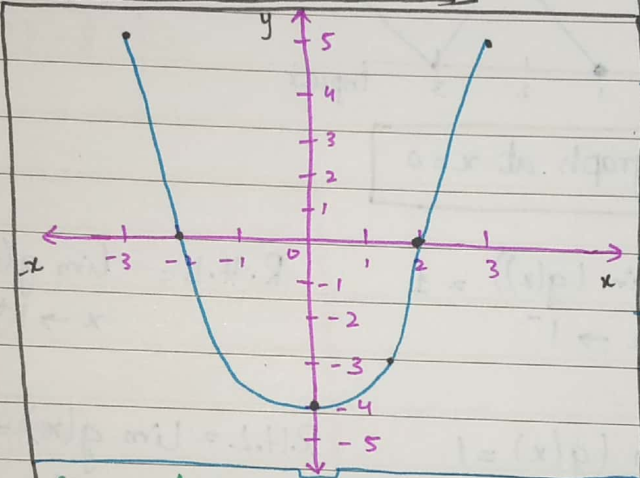
Range $(f(x)) = [0, 2)$

$y = f(x) = x^2 - 4$

Dom $(f(x)) = \mathbb{R}$

Independent	dependent
x	y
-3	5
-2	0
-1	-3
0	-4
1	-3
2	0
3	5

values for finding y



Graphical Representation of $x^2 - 4$

Limits of a Function

• When function is not define at some point and we are interested to study functional behaviour at nearby point.

$\lim_{x \to 1} f(x)$ = limiting value of $f(x)$ when $x \to 1$

$\frac{0}{0}$ is undeterminant form which doesn't have any specific value. it can be zero, can be one, can be two. undeterminant form can take any form

L.H.L $= \lim_{x \to 1^-} f(x)$

R.H.L $= \lim_{x \to 1^+} f(x)$

$\lim_{x \to 1} f(x)$ → Two sided limit

Theorem (existence of limit)

The limit $\lim_{x \to a} f(x)$ exist if

$\lim_{x \to a^-} f(x) =$ finite real number $= \lim_{x \to a^+} f(x)$

Further, $\lim_{x \to a} f(x)$ = finite real no

$f(1)$ = functional value at $x = 1$

$\lim_{x \to 0^-} f(x)$ = height / depth of graph when x is close to zero from left side

$\lim_{x \to 0^+} (f(x))$ = height / depth of graph when x is close to zero from right side

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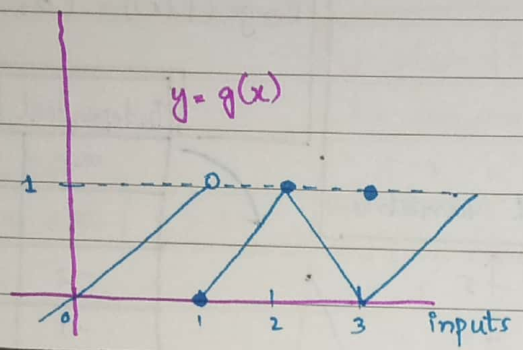
Exercise 2.1



1) $\lim_{x \to 1} g(x) \neq 0 = g(1)$

2) $\lim_{x \to 2} g(x) = 1 = g(2)$

c) $\lim_{x \to 3} g(x) \neq 1 = g(3)$

$\boxed{g(0) = \text{height | depth of graph at } x = 0}$

a) $\lim_{x \to 1} g(x)$ L.H.L $= \lim_{x \to 1^-} (g(x)) = 1$ R.H.L $= \lim_{x \to 1^+} g(x) = 0$

b) $\lim_{x \to 2} g(x)$ L.H.L $= \lim_{x \to 2^-} (g(x) = 1$ R.H.L $= \lim_{x \to 2^+} g(x) = 1 = 1$

As L.H.L = R.H.L so limit exist

Mathematical Calculation of Limits:

Exercise 2.2

Q1: $\lim_{x \to -7} (2x + 5)$

Solution:

$= \lim_{x \to -7} (2x) + \lim_{x \to -7} 5$

$= 2 \lim_{x \to -7} x + 5$

$= 2(-7) + 5$

$\lim_{x \to -7} (2x+5) = -9 \rightarrow$ limiting value.

Properties of limits:

a) $\lim_{x \to a} (\text{constant}) = \text{constant}$

b) $\lim_{x \to a} (k f(x)) = k \lim_{x \to a} f(x)$

c) $\lim_{x \to a} (f(x) \pm g(x)) = \lim_{x \to a} (f(x)) \pm \lim_{x \to a} (g(x))$

d) $\lim_{x \to a} x^n = \left(\lim_{x \to a} x\right)^n = a^n$

e) $\lim_{x \to a} (f(x) \cdot g(x)) = \left(\lim_{x \to a} f(x)\right) \cdot \left(\lim_{x \to a} g(x)\right)$

f) $\lim_{x \to a} \frac{(f(x))}{(g(x))} = \frac{\lim_{x \to a} (f(x))}{\lim_{x \to b} (g(x))}$

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Q5: $\lim_{x \to 6} 8(t-5)(t-7)$

$= 8\left[\lim_{t\to 6}(t-5) \cdot \lim_{t\to 7}(t-7)\right]$

$= 8[(6-5)\cdot(6-7)]$

$= 8(1)(-1)$

$\boxed{\lim_{t\to 6} 8(t-5)(t-7) = -8}$

Q10: $\lim_{y\to 2} \frac{y+2}{y^2+5y+6}$

$= \frac{\lim_{y\to 2}(y+2)}{\lim_{y\to 2}(y^2+5y+6)}$

$= \frac{\lim_{y\to 2} y + \lim_{y\to 2} 2}{\lim_{y\to 2} y^2 + 5\lim_{y\to 2} y + \lim_{y\to 2} 6}$

$= \frac{2+2}{4+10+6} = \frac{4}{20}$

$\boxed{\lim_{y\to 2} \frac{y+2}{y^2+5y+6} = \frac{1}{5}}$

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Limit of function of 0/0 form:

Question : 19 $\lim_{x\to5} \frac{x-5}{x^2-25}$

$= \frac{\lim_{x\to5}(x-5)}{\lim_{x\to5}(x^2-25)}$

$= \frac{5-5}{25-25} = 0/0$ form.

$\lim_{x\to5} \frac{x-5}{x^2-25} = \lim_{x\to5} \left(\frac{\cancel{x-5}}{(\cancel{x-5})(x+5)}\right)$

$= \lim_{x\to5} \frac{1}{x+5} \Rightarrow \frac{1}{5+5}$

$\boxed{\text{limit}_{x\to5} \frac{x-5}{x^2-25} = \frac{1}{10}}$

★ Important points:

- $\frac{0}{\text{non-zero}} = 0$
- $\frac{\text{non-zero}}{0} = \text{error/undefine} = \infty$
- $\frac{0}{0} \rightarrow$ undeterminant form
  it can take any value.

★ Rule for 0/0 form Type limits.

if we get 0/0 form in limits we try to factorize/rationalize both nominator and denominator and cancel those like term that make 0/0 form. and then apply the limit.

Extra Question

$\lim_{x\to5} \frac{x+5}{x^2-25} \Rightarrow \frac{\lim_{x\to5}(x+5)}{\lim_{x\to5}(x^2-25)}$

$= \frac{5+5}{25-25} = \frac{10}{0}$

$= \frac{10}{0}$ = undefined, limits doesn't exists.

$\lim_{x\to5} \frac{x-5}{x^2+25} \Rightarrow \frac{\lim_{x\to5}(x-5)}{\lim_{x\to5} x^2+25}$

$= \frac{5-5}{25+25} = \frac{0}{50} \Rightarrow 0$

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Q:24: $\lim_{t\to -1} \frac{t^2+3t+2}{t^2-t-2}$

$= \frac{\lim_{t\to -1}(t^2+3t+2)}{\lim_{t\to -1}(t^2-t-2)}$

$= \frac{(-1)^2+3(-1)+2}{(-1)^2-(-1)-2} = \frac{1-3+2}{1+1-2} = \frac{0}{0}$ form

$\lim_{t\to -1} \frac{t^2+3t+2}{t^2-t-2} = \lim_{t\to -1} \frac{(t+2)\cancel{(t+1)}}{(t-2)\cancel{(t+1)}}$

$\lim_{t\to -1} \frac{-1+2}{-1-2} = \frac{-1}{3}$

for making factors we use mid term break

Mid term break

$t^2+3t+2 = t^2+2t+t+2$

$= t(t+2)+1(t+2)$

$= (t+2)(t+1)$

$t^2-t-2 = t^2-2t+t-2$

$= t(t-2)+1(t-2)$

$= (t-2)(t+1)$

Limits where $x \to \pm\infty$

- for very large value of x $+\infty$
- for very small value of x $-\infty$

Example:

$\lim_{x\to\infty} \left(\frac{2x+3}{5x+4}\right)$

$= \frac{\lim_{x\to\infty}(2x+3)}{\lim_{x\to\infty}(5x+4)}$

$= \frac{2\lim_{x\to\infty} x + 3}{5\lim_{x\to\infty} x + 4}$

$= \frac{2(\infty)+3}{5(\infty)+4} = \frac{\infty}{\infty}$ form undermined form

formulas:

$a^2-b^2=(a-b)(a+b)$

$a^3-b^3=(a-b)(a^2+ab+b^2)$

$a^4-b^4=(a-b)(a^3-a^2b+ab^2+b^3)$

$a^{11}-a^{11}=(a-b)(a^{10}+a^9b+a^8b^2+a^7b^3$
$+a^6b^4+a^5b^5+a^4b^6+a^3b^7$
$+a^2b^8+a^1b^9+b^{10})$

$a^n-b^n=(a-b)(a^{n-1}+a^{n-2}b+a^{n-3}b^2$
$ab^{n-2}+b^{n-1})$

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$$\lim_{x\to\infty} \frac{2x+3}{5x+4} = \lim_{x\to\infty} \frac{x[2+3/x]}{x[5+4/x]}$$

or

$$= \frac{2+3\lim\limits_{x\to\infty} \frac{1}{x}}{5+4\lim\limits_{x\to\infty}\frac{1}{x}} = \frac{2+3(0)}{5+4(0)}$$

$$\lim_{x\to\infty} \frac{2x+3}{5x+4} = \frac{2}{5}$$

Rule $\frac{\infty}{\infty}$ form

- Take maximum power from x from nominator and denominator and cancel it.

- $\lim\limits_{x\to\pm\infty} \frac{1}{x} = 0$ | $\lim\limits_{x\to 0} \frac{1}{x} = \infty$
- $\lim\limits_{x\to\pm\infty} \frac{1}{x^2} = 0$ | $\lim\limits_{x\to 0} \frac{1}{x^2} = \infty$
- $\lim\limits_{x\to\pm\infty} \frac{1}{x^{1/2}} = 0$ | $\lim\limits_{x\to 0} \frac{1}{x^{1/2}} = \infty$
- $\lim\limits_{x\to\pm\infty} \frac{1}{x^{a}} = 0$ (a +ive number) | $\lim\limits_{x\to 0} \frac{1}{x^a} = \infty$

$$\lim_{x\to\infty} \frac{1}{x^2} = \lim_{x\to\infty} x^2 = \infty$$

Qs0 $\lim\limits_{x\to\infty} f(x) = \frac{3x+7}{x^2-2}$

$$\lim_{x\to\infty} \frac{3x+7}{x^2-2} \Rightarrow \frac{3(\infty)+7}{(\infty)^2-2} = \frac{\infty}{\infty}$$

$$\lim_{x\to\infty} \frac{3x+7}{x^2-2} = \frac{x(3+7/x)}{x^2(1-2/x^2)} = \left(\lim_{x\to\infty} \frac{1}{x}\right)\left(\lim_{x\to\infty} \frac{3+7/x}{1-2/x^2}\right)$$

$$= \frac{3+7\lim 1/x}{1-2\lim 1/x^2} = 0(3) = 0$$

- $\frac{x^{low}}{x^{high}} = 0$
- $\frac{x^1}{x^1}$ = constant non-zero
- $\frac{x^{high}}{x^{low}} = \infty$

Extra

$$\lim_{x\to\infty} \frac{x^4+x^3-1}{x^3+2x} = \frac{\infty^4+\infty^4-1}{\infty^3+2(\infty)} = \frac{\infty}{\infty} \text{ form}$$

$$\lim_{x\to\infty} \frac{x^4+x^3-1}{x^3+2x} = \lim_{x\to\infty} \frac{x^4(1+1/x-1/x^4)}{x^3\left[1+\frac{2}{x^2}\right]}$$

$$= \left(\lim_{x\to\infty} x\right)\left(\frac{1+\lim\limits_{x\to\infty}\frac{1}{x} - \lim\limits_{x\to\infty}\frac{1}{x^4}}{1+2\lim\limits_{x\to\infty}\frac{1}{x^2}}\right)$$

$$= \infty(1)$$

$$= \infty \text{ (limit doesnot exist)}$$

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42. $\lim_{x \to \infty} \dfrac{3 - 2/x}{4 + \sqrt{2}/x^2} = \dfrac{3 - 2\lim_{x\to\infty} \cancel{1/x}^{\,0}}{4 + \sqrt{2}\lim_{x\to\infty} \cancel{1/x^2}^{\,0}} = \dfrac{3}{4}$

Q25: $\lim_{x \to 0} \dfrac{\tan 2x}{x}$

$\lim_{x\to 0} \dfrac{\tan 2x}{x} = \lim_{x\to 0} \dfrac{\frac{\sin 2x}{\cos 2x}}{x}$

$= \lim_{x\to 0} \dfrac{\sin 2x}{x} \cdot \dfrac{1}{\cos 2x}$

$= \left(\lim_{x\to 0} \dfrac{\sin 2x}{2x} \cdot 2\right)\left(\lim_{x\to 0} \dfrac{1}{\cos 2x}\right)$

$= 2\left(\lim_{x\to 0} \cancel{\dfrac{\sin 2x}{2x}}^{\,1}\right)\left(\lim \cancel{\dfrac{1}{\cos 2(0)}}^{\,1}\right)$

$= 2$

Application of Sandwich theorm

$\lim_{\boxed{x\to 0}} \dfrac{\sin x}{x} = 1$

1. must be same
2. limit apporach to 0
3. direct limit produce 0/0 form

• convert all function in sin, cos. form

$\cos 0 = 1$

Q36. $\lim_{y\to 0} \dfrac{\sin 3y \cot 5y}{y \cot 4y} = \lim_{y\to 0} \dfrac{\sin 3y}{y} \cdot \dfrac{\cos 5y}{\sin 5y} \cdot \dfrac{\sin 4y}{\cos 4y}$

$= 3\left(\lim_{y\to 0} \dfrac{\sin 3y}{3y}\right) \cdot \left(\lim_{y\to 0} \dfrac{5y}{\sin 5y}\right) \cdot \left(\lim_{y\to 0} \dfrac{\sin 4y}{4y}\right)\left(\lim_{y\to 0} \dfrac{\cos 5y}{\cos 4y}\right) \dfrac{4\cancel{y}}{5\cancel{y}}$

$= \dfrac{12}{5}(1)(1)(1)(1)$

$= 12/5$

Q31 = $\lim_{t\to 0} \dfrac{\sin(1 - \cos t)}{(1 - \cos t)}$

Suppose $x = 1 - \cos t$

$t \to 0, \quad x \to 0$

$\lim_{t\to 0} \dfrac{\sin(1 - \cos t)}{(1 - \cos t)} = \lim_{x\to 0} \dfrac{\sin x}{x} = 1$

• $\dfrac{\sin(0)}{0} = \dfrac{0}{0}$ form

$\sin(0) = 0$

$\cos(0) = 1$

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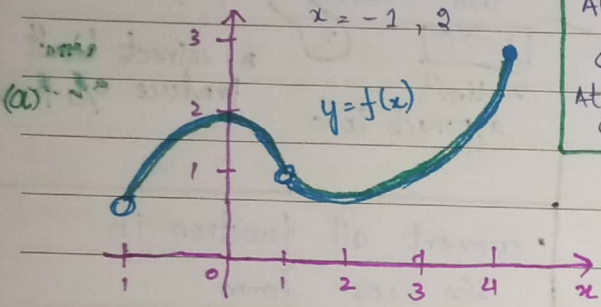
Continuity :

Continuity of a function at a point (from graph)

is not continous at $x = -1, 2$

(a)



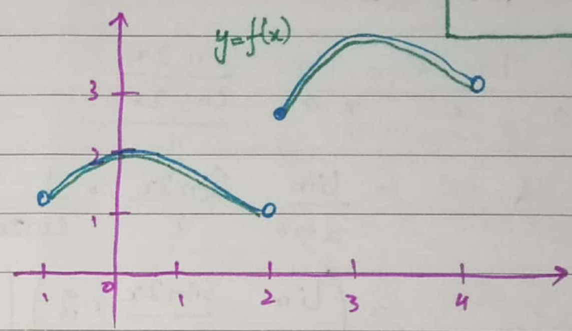
At $x = 1$

C1; X

At $x = 2$

C1 ✓

(b)

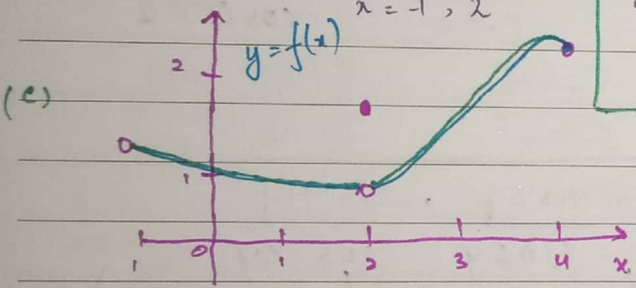


At $x = 2$

C1: ✓

is not continus at $x = -1, 2$

(c)

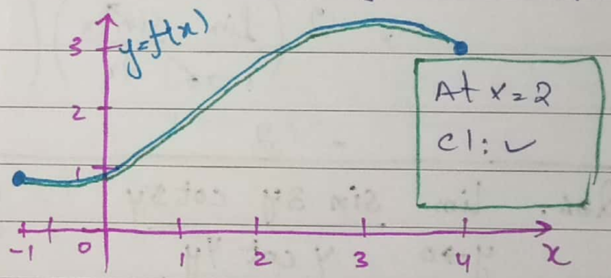


At $x = 2$

C: ✓

is continous every where at all point

(d)



At $x=2$

C1: ✓

Continuity Test: A function $y = f(x)$ is said to be continous at $x = a$ if it satisfies following 3-conditions.

(Mathematically,)

C1: function should be define at $x = a$ i.e $f(a)$ is finite real number.

Graphically, this means that graph of a function have no hole.

C2: $\lim_{x \to a} f(x)$ exist i.e. $\lim_{x \to a^-} f(x) =$ finite real number $= \lim_{x \to a^+} f(x)$

(hight & depth same) Graphically

Left hand limit

Right hand limit

C3: functional value at $x = a$ should be equal to limiting value $x \to a$

$$f(a) = \lim_{x \to a} f(x)$$

(functional) (Limiting Value)

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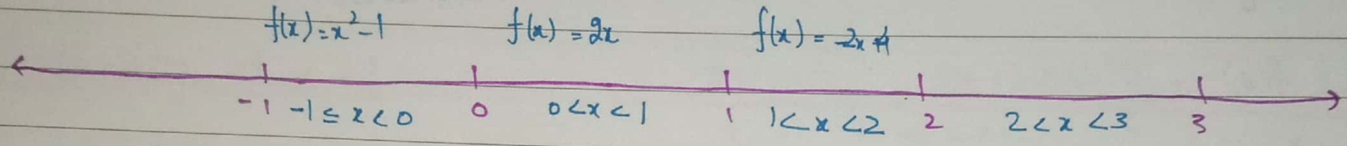
Discuss continuity of function $f(x)$

$$f(x) = \begin{cases} x^2 - 1, & -1 \le x < 0 \\ 2x, & 0 < x < 1 \\ 1, & x = 1 \\ -2x + 4, & 1 < x < 2 \\ 0, & 2 < x < 3 \end{cases}$$

where There is equal sign it means that function is continous at that point.



Continuity at $x = -1$

C1: $f(-1) = (-1)^2 - 1 = 0$ C1 ✓

C2: $\lim\limits_{x \to -1^+} f(x) = \lim\limits_{x \to -1^-} (x^2 - 1) = 0$ C2 ✓

C3: $f(-1) = 0 = \lim\limits_{x \to -1}$ C3 ✓

function is continous at $x = -1$

Continuity at $x = 1$

C1: $f(1) = 1$

C2: $\underset{x \to 1}{\text{L.H.L}} = \lim\limits_{x \to 1^-} f(x) = \lim\limits_{x \to 1} 2x = 2$

C3: $\lim\limits_{x \to 1^+} f(x) = \lim\limits_{x \to 1^+} (-2x + 4) = -2 + 4 = 2$

C3: $f(1) = \lim\limits_{x \to 1} f(x)$

$1 \ne 2$

Continuity at $x = 0$

C1: $f(0)$ = not define so function is not continous

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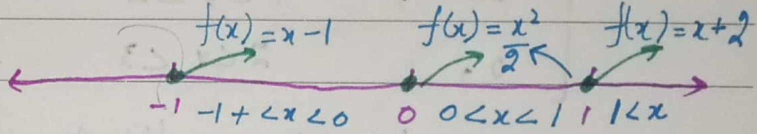
$$f(x) = \begin{cases} x-1 & \text{if } -1 \leq x < 0 \\ x^2/2 & 0 \leq x \leq 1 \\ x+2 & 1 \leq 0 \end{cases}$$



Continuity At $x=0$

$C_1 = f(0) = \frac{0^2}{2} = 0$

C_2 : Left hand Limit $= \lim_{x \to 0^-} f(x) = \lim_{x \to 0} (x-1) \Rightarrow -1$

Right hand limit $= \lim_{x \to 0^+} f(x) = \lim_{x \to 0} \frac{x^2}{2} = 0$

L.H.L $\neq$ R.H.L

function is discontinuous at 0.

Continuity At $x = 1$

$C_1 = f(1) = \frac{x^2}{2} = \frac{1}{2}$

$C_2 =$ L.H.L $= \lim_{x \to 1^-} \frac{x^2}{2} = \frac{1}{2}$

R.H.L $= \lim_{x \to 1^+} x+2 = 3$

L.H.L $\neq$ R.H.S

function is discontinues.

Continuity at $x = -1$

$C_1 = f(-1) = \cancel{x} - 1$

$= -1-1 = -2$

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$C_2 =$ L.H.L $\lim_{x \to 1^-} \cancel{(x-1)} = \cancel{2} -2$

R.H.L $\cancel{\lim_{x \to 1} (}$

$$f(x) = \begin{cases} x^3 & \text{if } x < -1 \\ ax+b & \text{if } -1 \le x < 1 \\ x^2+2 & \text{if } 1 \le x \end{cases}$$

<u>Continuity at $x = -1$</u>

$f(x) = x^3$; $ax+b = f(x)$; $x^2+2 = f(x)$

$x < -1$; -1 ; $-1 < x < 1$; 1 ; $1 < x$

$C_1 = f(-1) = ax+b \Rightarrow -a+b$

$C_2 =$ L.H.L $= \lim_{x \to -1^-} x^3 = -1$

R.H.L $\lim_{x \to -1^+} ax+b \Rightarrow -a+b$

Given that $f(x)$ is continous at $x = -2$

L.H.L = R.H.L

$-1 = -a+b \Rightarrow \cancel{a-b} = 1$ $\quad a-1 = b$ — (1)

<u>Continuity at $x = 1$</u>

$C_1 = f(1) = x^2+2 = 3$

$C_2 =$ L.H.L $= \lim_{x \to 1^-} ax+b = a+b$

R.H.L $= \lim_{x \to 1^+} x^2+2 = 1+2 = 3$

Given that $f(x)$ is continous at $x = 1$

$a+b = 3$ — (2)

$(a)+(a-1) = 3$

$a+a = 3+1$

$2a = 4$

$\boxed{a = 2}$

$-1 = -2+b$

$-1+2 = b$

$\boxed{b = 1}$

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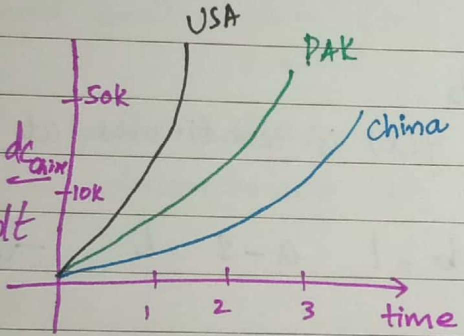
Derivative
~~Determinants~~

rate = how fast?
How slow?

- Whenever we change the independent quantity there whold be a change in independent quantity
- Rate of change of dependent variable with respect to independent variable
- whenever 2 or more quantity related to each other we define function
- when we want to see How some will change with respect to other quantity then we have to calcute the detivative.
- Derivative of function represents how much a function is sensitive to its input

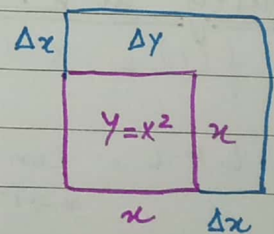
Mathematically,

$$\frac{dc_{USA}}{dt} > \frac{dc_{PAK}}{dt} > \frac{dc_{CHINA}}{dt}$$



From graph we can see that no. of covid cases for USA is more sensitive ~~to~~ as compaired to Pakistan and china

Mathematical defination of derivative:



$y = f(x)$

dependent ← y, x → independent

$y + \Delta y = f(x + \Delta x)$ (change in dependent: Δy; chang in independent: Δx) $\Rightarration$ $\dfrac{\Delta y}{\Delta x} = \dfrac{f(x+\Delta x) - f(x)}{\Delta x}$ } ration, average change, we $\div \Delta x$ on b.s

For very small

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} \Rightarrow \frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

intstantanious change

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~~Technignique~~

Techniques of Derivative

Rule for constant

$$\frac{d}{dx}(2f(x)) \Rightarrow 2\frac{d}{dx}(fx)$$

- constant outside of derivative

Sum rule

$$\frac{d(f(x) \pm g(x))}{dx} = \frac{df}{dx} \pm \frac{dg}{dx}$$

$\frac{d}{dx}(\text{constant}) = 0$

$\frac{d}{dt}(\text{constant}) = 0$

$\frac{d}{dx} \rightarrow$ Derivative w.r.t x

$\frac{d}{dt} \rightarrow$ Derivative w.r.t t

$\frac{d(u)}{dt}$ –

Product rule:

$$\frac{d}{dx}(f(x)\,g(x)) = f(x)\frac{dg}{dx} + g(x)\frac{df}{dx}$$

For 3 terms

$$\frac{d}{dx}(f(x).g(x).h(x)) = f(x)(g(x).\frac{dh}{dx} + f(x).\frac{dg}{dx}.h(x) . \frac{df}{dx}(gx).h(x)$$

Quoient Rule:

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x).\frac{df}{dx} - f(x).\frac{dg}{dx}}{(g(x))^2}$$

Power Rule of derivative/differentiation:

Power function = $(\text{Variable})^{\text{constant}}$

$f(x) = x^{a}$ $a \rightarrow$ constant

$$\frac{df}{dx} = a.x^{a-1}$$

$y = x^{x}$ ✗ wrong

$\frac{dy}{dx} = x.x^{x-1}$

Shift constant power to the start and reduce the power by 1

power can be positive or negative.

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Example: 1 $f(x) = \frac{1}{x} = x^{-1} = -1 \cdot x^{-1-1} = -x^{-2} \Rightarrow -\frac{1}{x^2}$

Example: 2 $f(x) = \sqrt{x} = x^{1/2}$ $\frac{df}{dx} = \frac{1}{2} x^{\frac{1}{2}-1} = \frac{1}{2\sqrt{x}}$

Example 3 $f(x) \frac{1}{\sqrt[3]{x}} = x^{-1/3}$

$\boxed{\text{Generalized Power Rule:}}$

$f(x) = (\text{function})^{\text{constant}}$

$f(x) = (g(x))^a$

$\frac{df}{dx} = a(g(x))^{a-1} \frac{dg}{dx}$

$\frac{df}{dx} = \text{constant}(\text{function})^{\text{constant}-1} \cdot \frac{d}{dx}(\text{constant})$

$\boxed{\text{Example}}$

- $f(x) = (2x^2 + x - 1)^{\frac{1}{2}}$

~~$\frac{d}{dx}$ $y = (2x^2+x-1)^{\frac{1}{2}}$~~

~~$\frac{dy}{dx} = \frac{1}{2}(2x^2+x-1)^{\frac{1}{2}-1} \cdot \frac{d}{dx}(2x^2+x-1)$~~

~~$\frac{dy}{dx} = \frac{1}{2}$ $2\frac{dx^2}{dx} + \frac{dx}{dx} - 1$~~

$\frac{df}{dx} = \frac{d}{dx}(2x^2 + x - 1)^{\frac{1}{2}}$

$= \frac{1}{2}(2x^2+x-1)^{\frac{1}{2}-1} \cdot \frac{d}{dx}(2x^2+x-1)$

$= \frac{1}{2}(2x^2+x-1)^{-\frac{1}{2}} \cdot [\cancel{4x} + \frac{d}{dx}(2x^2) + \frac{d}{dx}(x) - \frac{d}{dx}(1))$

$= \frac{1}{2}(2x^2+x-1)^{-\frac{1}{2}} \cdot [4x+1]$

$= \frac{4x+1}{2\sqrt{2x^2+x-1}}$

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$$f(x) = \sqrt[3]{\frac{2x^2+1}{x-1}}$$

$$\left(\frac{2x^2+1}{x-1}\right)^{\frac{1}{3}}$$

$$\frac{\sqrt{(x-1)}^{x}}{x-1}$$

$$\frac{df}{dx} = \frac{d}{dx}\left(\frac{2x^2+1}{x-1}\right)^{\frac{1}{3}}$$

$$= \frac{1}{3}\left(\frac{2x^2+1}{x-1}\right)^{\frac{1}{3}-1} \cdot \frac{d}{dx}\left(\frac{2x^2+1}{x-1}\right)$$

$$= \frac{1}{3}\left(\frac{2x^2+1}{x-1}\right)^{-\frac{2}{3}} \cdot \frac{(x-1)\frac{d(2x^2+1)}{dx} - (2x^2+1)\frac{d}{dx}(x-1)}{(x-1)^2}$$

$$= \frac{1}{3}\left(\frac{2x^2+1}{x-1}\right)^{-\frac{2}{3}} \cdot \frac{(x-1)(4x) - (2x^2+1)(1)}{(x-1)^2}$$

$$= \frac{1}{3}\left(\frac{2x^2+1}{x-1}\right)^{-\frac{2}{3}} \cdot \frac{4x^2-4x-2x^2-1}{(x-1)^2}$$

$$= \frac{1}{3}\left(\frac{x-1}{2x^2+1}\right)^{\frac{2}{3}} \cdot \frac{2x^2-4x-1}{(x-1)^2}$$

Derivative of exponential Rule:

Exponential function = $(\text{constant})^{\text{variable}}$

always positive and $\neq 1$

$$\frac{d(E.F)}{dx} = (\text{constant})^{\text{variable}} \cdot \ln(\text{constant})$$

$y = 2^x$
$y = e^x$
$y = 10^x$
$y = \left(\frac{1}{2}\right)^x$

- $y = 2^x$, $\frac{dy}{dx} = 2^x \cdot \ln 2$
- $y = a^x$, $\frac{dy}{dx} = a^x \cdot \ln a$
- $y = e^x$, $y = e^x \cdot \ln e$

- $F(x) = a^x = \frac{df}{dx} = a^x \cdot \ln a$

Generalized

$$f(x) = a^{g(x)}$$

$$\frac{df}{dx} = a^{g(x)} \cdot \ln a \cdot \frac{d}{dx}(g(x))$$

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Generalized Exponential Rule

$f(x) = (\text{constant})^{\text{function}}$

$\frac{d f(x)}{d} = (\text{constant})^{\text{function}} \cdot \ln(\text{constant}) \cdot \frac{d}{dx}(\text{variable})$

$f(x) = 2^{x^2+1}$

$\frac{df}{dx} = 2^{x^2+1} \cdot \ln 2 \cdot \frac{d}{dx}(x^2+1)$

Derivative of log function:

$y = \log_a x$

$\frac{dy}{dx} = \frac{1}{\ln a x}$

$y = \log_2 x$

$\frac{dy}{dx} = \frac{1}{x} \frac{1}{\ln 2}$

$y = \ln x$

$\frac{dy}{dx} = \frac{1}{x}$

Rules:

one over angel.

Generalized log function

$y = \ln(\cancel{x^2+x}\ x^2+1)$

$\frac{dy}{dx} = \frac{1}{x^2+1} \cdot \frac{1}{\cancel{\ln e}} 1 \cdot \frac{d}{dx}(x^2+1)$

$$\boxed{\frac{dy}{dx} = \frac{d}{dx}\left(\log_a(\text{angle})\right) = \left(\frac{1}{\ln a} \cdot \frac{1}{\text{angle}} \cdot \frac{d}{dx}(\text{angle})\right)}$$

$f(x) = 2^x$

$f^{-1}(x) = \log_2 x$

$f(x) = e^x$ natural log

$f^{-1}(x) = \log_e x = \ln x$

$f(x) = 10^x$

$f^{-1}(x) = \log_{10} x$

$= \log x$ common log

Derivative of trignometric function:

$$\boxed{\frac{d}{dx}(\sin x) = \cos x}$$

Tip:

$s \longleftrightarrow \cos$

$\tan \longleftrightarrow \sec x$

$\cot \longleftrightarrow \operatorname{cosec} x$

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- $\frac{d}{dx}(\cos x) = -\sin x$

- $\frac{d}{dx}(\tan x) = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right) = \frac{\cos x(\cos x) - (\sin x)(-\sin x)}{\cos^2 x}$

$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$

$= \frac{1}{\cos^2 x} = \sec^2 x$

- $\frac{d}{dx}(\tan x) = \sec x \cdot \sec x = \sec^2 x$

- $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

- $\frac{d}{dx}(\sec x) = \sec x \cdot \tan x$

- $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

trignometry Generalized

- $\frac{d(\sin(f(x)))}{dx} = \cos(f(x))\frac{df}{dx}$

- $\frac{d}{dx}(\cos(f(x))) = -\sin(f(x)) \cdot \frac{df}{dx}$

- $\frac{d}{dx}(\tan(f(x))) = \sec^2(f(x)) \cdot \frac{df}{dx}$

- $\frac{d}{dx}(\cot(f(x))) = -\operatorname{cosec}^2(f(x)) \cdot \frac{df}{dx}$

- $\frac{d}{dx}(\sec(f(x))) = \sec(f(x))\tan(f x) \cdot \frac{df}{dx}$

- $\frac{d}{dx}(\operatorname{cosec}(f(x))) = \operatorname{cosec}(f(x)) \cdot \cot(f(x)) \cdot \frac{df}{dx}$

Tips 1

- derivative of all the function that starts from c are -ive

- Tip 2:

$\frac{d}{dx}\left(\frac{1}{\cos x}\right) = \frac{d}{dx}\left((\cos x)^{-1}\right)$

$= -1(\cos x)^{-2} - \sin x$

$= \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x}$

$\frac{d}{dx}(\sec x) = \tan x - \sec x$

$\frac{d}{dx}(\sec(\ln x))$

$= \sec(\ln x) \cdot \tan(\ln x) \cdot \frac{d}{dx}(\ln x)$

$\frac{d}{dx}\left(\tan(2^{x^2})\right)$

$= \sec^2(2^{x^2}) \cdot \frac{d}{dx}(2^{x^2})$

$= \sec^2(2^{x^2}) \cdot \left(2^{x^2} \cdot \ln 2 \cdot \frac{d}{dx}(x^2)\right)$

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Derivative of inverse trigonometric Function:

- $(y = \sin^{-1}x) = \text{arc} \sin(x)$

$\sin y = x$

Diff w.r.t x

$\cos y \cdot \frac{dy}{dx} = 1$

$\frac{dy}{dx} = \frac{1}{\cos y}$

$\frac{dy}{dx} = \frac{1}{\sqrt{1-\sin^2 y}}$

$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$

- $y = \tan^{-1}x \quad \frac{dy}{dx} = ?$

$\tan y = x$

Diff w.r.t y

$\sec^2 y \cdot \frac{dy}{dx} = 1$

$\frac{dy}{dx} = \frac{1}{\sec^2 y}$

$= \frac{1}{1+\tan^2 y}$

$\frac{dy}{dx} = (\tan^{-1}x) = \frac{1}{1+x^2}$

Formulas:

- $\frac{d}{dx}(\sin^{-1}(\text{angle})) = \frac{1}{\sqrt{1-(\text{angle})^2}} \cdot \frac{d}{dx}(\text{angle})$

- $\frac{d}{dx}(\tan^{-1}(\text{angle})) = \frac{1}{1+(\text{angle})^2} \cdot \frac{d}{dx}(\text{angle})$

$\therefore \sin^2 y + \tan^2 y = 1$

$\cos^2 y = 1 - \sin^2 y$

$\boxed{\cos y = \sqrt{1-\sin^2 y}}$

- $\frac{d}{dx}(\sec^{-1}(\text{angle})) = \frac{1}{\text{angle}\sqrt{\text{angle}^2-1}} \cdot \frac{d}{dx}(\text{angle})$

- $\frac{d}{dx}(\cos^{-1}(\text{angle})) = \frac{-1}{\sqrt{1+(\text{angle})^2}} \cdot \frac{d}{dx}(\text{angle})$

- $\frac{d}{dx}(\cot^{-1}(\text{angle})) = -\frac{1}{1+(\text{angle})^2} \cdot \frac{d}{dx}(\text{angle})$

- $\frac{d}{dx}(\text{cosec}^{-1}(\text{angle})) = -\frac{1}{\text{angle}\sqrt{(\text{angle})^2-1}} \cdot \frac{d}{dx}(\text{angle})$

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Find $\frac{dy}{dx}$ if $y = \left(\ln\left(\sin^{-1}\left(\frac{2^x + \tan x}{x^2+\sec x}\right)\right)\right)^{2/3}$

Diff w.r.t x :

$$\frac{dy}{dx} = \frac{3}{2}\left(\ln\left(\sin^{-1}\left(\frac{2^x + \tan x}{x^2+\sec x}\right)\right)\right)^{\frac{3}{2}-1} \cdot \frac{d}{dx}\left[\ln\left(\sin^{-1}\left(\frac{2^x+\tan x}{x^2+\sec x}\right)\right)\right]$$

$$\cdot \frac{1}{\sin^{-1}\left(\frac{2^x+\tan x}{x^2+\sec x}\right)} \cdot \frac{1}{\cancel{\ln e}} \frac{d}{dx}\left(\sin^{-1}\left(\frac{2^x+\tan x}{x^2+\sec x}\right)\right)$$

Now

$$\frac{d}{dx}\left(\sin^{-1}\left(\frac{2^x+\tan x}{x^2+\sec x}\right)\right) = \frac{1}{\sqrt{1-\left(\frac{2x+\tan x}{x^2+\sec x}\right)^2}} \cdot \frac{d}{dx}\left(\frac{2^x+\tan x}{x^2+\sec x}\right)$$

$$= \frac{x^2+\sec x}{\sqrt{(x^2+\sec x)^2-(2^x+\tan x)^2}} \cdot \left[\frac{x^2+\sec \cdot \frac{d}{d}(2^x+\tan x)}{(x^2+\sec x)^2}\right.$$

$$\left[\frac{(x^2+\sec)\left[2^x\ln 2+\sec^2 x\right]-(2^x+\tan x)\left(2^x+\cancel{\sec x}\tan x\right)}{(x^2+\sec x)^2}\right]$$

① Find $\frac{dy}{dx}$ if $y = \log_{10}\sqrt{2^{x^2}+\sin^{-1}x+x^2\sin x}$

Power

② $y = \cancel{2}^{x^2+\ln(x+1)} \cdot \sin^{-1}x + \ln(\sqrt{x}+1)$

③ $y = \ln(x^2+\sqrt{x}+1) \cdot \sin^{-1}\left(\frac{x+1}{x-1}\right)$

Ans 1:

$Y = \log_{10}\sqrt{2^{x^2}+\sin^{-1}x+x^2\sin x}$

4) $y = (\sin x)^{\cos x} + (\tan x)^{\sec x}$

Date — function

Use of log to find Derivative

1) Derivative of a function having variable Power:

Example:

$y = 2^x$ find $\frac{dy}{dx}$

$\ln y = \ln 2^x$

$\ln y = x \ln 2$

Diff w.r.t. x

$\frac{1}{y} \cdot \frac{dy}{dx} = \ln 2$

$\frac{dy}{dx} = y \ln 2$

$$\boxed{\frac{dy}{dx} = 2^x \ln 2}$$

Properties of log

Power rule

- $\ln M^N = N \ln M$
- $\ln(MN) = \ln(M) + \ln(N)$
- $\ln\left(\frac{M}{N}\right) = \ln(M) - \ln(N)$

Tip:

- if we have variable quantity on power we use log on both side.

Example:

$y = x^x$

$\ln y = \ln x^x$

$\ln y = x \ln x$

Diff w.r.t x

$\frac{1}{y}\frac{dy}{dx} = \frac{d}{dx}(x \ln x)$

$= x \cdot \frac{1}{x} + \ln x$

$\frac{1}{y}\frac{dy}{dx} = 1 + \ln x$

$\frac{dy}{dx} = (y)(1 + \ln x)$

$$\boxed{\frac{dy}{dx} = (x^x)(1 + \ln x)}$$

Example $y = \sin x^{\cos x}$

$\ln y = \ln \sin x^{\cos x}$

$\ln y = \cos x \ln \sin x$

Diff w.r.t x

$\frac{1}{y}\frac{dy}{dx} = \frac{d}{dx}(\cos x \ln \sin x)$

$\frac{1}{y}\frac{dy}{dx} = \cos x \frac{d}{dx}(\ln \sin x) + \frac{d}{dx}(\cos x)(\ln \sin x)$

$= \cos x \cdot \frac{1}{\sin x} \cdot \cos x + \ln(\sin x)(-\sin x)$

$\frac{1}{y}\frac{dy}{dx} = \cot x \cdot \cos x - \sin x \ln(\sin x)$

$$\boxed{\frac{dy}{dx} = (\sin x)^{\cos x}\left[\cot x \cos x - \sin x (\ln \sin x)\right]}$$

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$$\frac{d}{dy}(\ln y) = \frac{1}{y}$$

$$\frac{d}{dx}(\ln y) = \frac{1}{y}\frac{dy}{dx}$$

$$y = \frac{(2x+e^x)^{3/2}\,(\sin x + x^2)^{\frac{1}{2}}}{(x^2+\sqrt{x}+1)^5\,(\sin^{-1}x + x^x)}$$

Solution

$$\ln y = \ln\left(\frac{(2x+e^x)^{3/2}(\sin x + x^2)^{\frac{1}{2}}}{(x^2+\sqrt{x}+1)^5(\sin^{-1}x + x^x)}\right)$$

$$= \ln(2x+e^x)^{3/2} + \ln(\sin x + x^2)^{\frac{1}{2}} - \ln(x^2+\sqrt{x}+1)^5 - \ln(\sin^{-1}x + x^x)$$

$$= \frac{3}{2}\ln(2x+e^x) + \frac{1}{2}\ln(\sin x + x^2) - 5\ln(x^2+\sqrt{x}+1) - \ln(\sin^{-1}x + x^x)$$

$$= \frac{3}{2}\,\frac{1}{2x+e^x}\,\frac{d}{dx}(2x+e^x) + \frac{1}{2}\,\frac{1}{\sin x + x^2}\cdot\frac{d}{dx}(\sin x + x^2) - 5\,\frac{1}{x^2+\sqrt{x}+1}\,\frac{d}{dx}(x^2+\sqrt{x}+1)$$

$$- \frac{1}{\sin^{-1}x + x^x}\cdot\frac{d}{dx}(\sin^{-1}x + x^x)$$

$$\frac{1}{y}\frac{dy}{dx} = \frac{3}{2}\,\frac{2+e^x}{2x+e^x} + \frac{1}{2}\,\frac{\cos x + 2x}{\sin x + x^2} - 5\,\frac{2x+\frac{1}{2\sqrt{x}}}{x^2+\sqrt{x}+1} - \frac{\frac{1}{\sqrt{1-x^2}} + x^x(1+\ln x)}{\sin^{-1}x + x^x}$$

$$= y\Big(\rule{8em}{0.4pt}\Big)$$

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Chain Rule is used to calculate the derivative of composite function $\rightarrow$ function inside another function.

Chain Rule of Differentiation:-

Composite function:

$$g(f(x)) = g\circ f(x)$$

$$\begin{tikzcd} x \arrow[r, "f"] \arrow[dr, "g\circ f"'] & y \arrow[d, "g"] \\ & z \end{tikzcd}$$

$$\begin{tikzcd} x \arrow[r, "g"] \arrow[dr, "f\circ g"'] & Y \arrow[d, "f"] \\ & z \end{tikzcd}$$

$$\underbrace{f\circ g(x) = f(g(x))}_{\text{composite function}}$$

Simple function

$f(x) = x^a$

$f(x) = \ln x$

$f(x) = 2^x$

Composite function

$f(x) = (\sin x)^a$

$g(x) = \sin x$

$f(x) = x^a$

$\downarrow$

$f(g(x)) = (\sin x)^a$

$f(x) = \ln(x^2+1)$

outer function

inner function

Generalized composite function:

$y = f(g(x))$ $\quad \dfrac{dy}{dx} = ?$

Suppose $u = g(x)$

$y = f(u)$ with $u = g(x)$

$$\frac{dy}{dx} = \frac{d}{dx}(f(u)) = \frac{df}{du}\cdot\frac{du}{dx}$$

$$\boxed{\frac{dy}{dx} = \frac{df}{du}\cdot\frac{du}{dx} = \frac{dy}{du}\,\frac{du}{dx}}$$

Example:

$y = \ln(x^2+1)$

Suppose $u = (x^2+1)$

$y = \ln u$

Diff wrt x

$$\frac{dy}{dx} = \frac{dy}{du}\,\frac{du}{dx} = \frac{1}{u}\cdot\frac{d}{dx}(x^2+1)$$

$$\frac{dy}{dx} = \frac{1}{x^2+1}\cdot\frac{d}{dx}(x^2+1)$$

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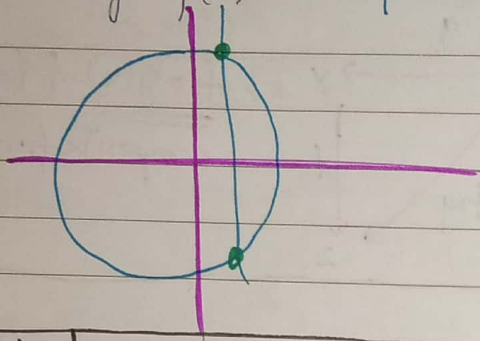
Parametric Differentiation:

$y = f(x) \rightarrow$ explicit function



For circle: $x^2 + y^2 = a^2$

circle with centre $(0,0)$, radius $= a$

$$\begin{cases} x = a\cos\theta \\ y = a\sin\theta \end{cases}$$ Parametric equation of circle

$$\boxed{\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}}$$ formula of parametric equation

$$\frac{dy}{dx} = \frac{dy}{dx} \cdot \frac{dx}{d\theta}$$

Example:

$y = \sin^{-1}(x^2+1)$

Suppose $y = \sin^{-1}(x^2+1)$ w.r.t $u = \ln(x^2+1)$

$$\frac{dy}{du} = \frac{dy/dx}{du/dx}$$

Implicit Differentiation:

Implicit Relation: $f(x,y) = c$ ← constant

$$\sin^{-1}(x^2+y^2) + \ln\left(\frac{x}{y}\right) + 2^{y/x} = 1$$

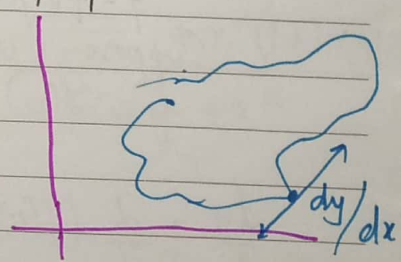
Diff with respect to x

$$\cancel{\frac{1}{\sqrt{1-(x^2+y^2)}}\frac{d}{dx}(x^2+y^2) + \frac{y}{x}\frac{d}{dx}\left(\frac{x}{y}\right) + 2^{y/x}\ln 2}$$

$$\frac{1}{\sqrt{1-(x^2+y^2)}}\frac{d}{dx}(x^2+y^2) + \frac{1}{x} - \frac{1}{y}\cdot\frac{dy}{dx} + 2^{y/x}\cdot\ln 2 \cdot \frac{d}{dx}\left(\frac{y}{x}\right) = 0$$

$$\frac{1}{\sqrt{1-(x^2+y^2)}}\left[2x + 2y\frac{dy}{dx}\right] + \frac{1}{x} - \frac{1}{y}\frac{dy}{dx} + 2^{y/x}\cdot\ln 2\ \frac{x\frac{dy}{dx} - y}{x^2} = 0$$

Graph



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$$\frac{dy}{dx}\left[\frac{2y}{\sqrt{1-(x^2+y^2)^2}}-\frac{1}{y}+\frac{2^{y/x}\ln 2x}{x^2}\right]=\frac{-2x}{\sqrt{1-(x^2+y^2)}}-\frac{1}{x}+\frac{2^{y/x}\ln 2\cdot y}{x^2}$$

$$\frac{dy}{dx}=\frac{\dfrac{-2x}{\sqrt{1-(x^2+y^2)}}-\dfrac{1}{x}+\dfrac{2^{y/x}\ln 2\cdot y}{x^2}}{\left[\dfrac{2y}{\sqrt{1-(x^2+y^2)^2}}-\dfrac{1}{y}+\dfrac{2^{y/x}\ln 2x}{x^2}\right]}$$

Easy way to solve these types of Question:

$f(x, y)=c \rightarrow$ implicit Relation.

$$df=\frac{\partial f}{\partial x}dx+\frac{\partial f}{\partial y}dy=d(c)$$

$$\frac{\partial f}{\partial x}dx+\frac{\partial f}{\partial y}dy=0$$

$$\frac{\partial f}{\partial y}dy=-\frac{\partial f}{\partial x}dx$$

$$\boxed{\frac{dy}{dx}=\frac{\partial f/\partial x}{\partial f/\partial y}}$$

$\frac{\partial f}{\partial x}=$ Derivative w.r.t x while keeping y as constant

$\frac{\partial f}{\partial y}=$ Derivative w.r.t y while keeping x as constant

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$$f(x,y) = \sin^{-1}(x^2+y^2) + \ln\left(\frac{x}{y}\right) + 2^{y/x}$$

$$\frac{\partial f}{\partial x} = \frac{2x}{\sqrt{1-(x^2+y^2)^2}} + \frac{1}{x} + 2^{y/x} \cdot \ln 2 \cdot y \cdot \left(\frac{-1}{x^2}\right)$$

$$\frac{\partial f}{\partial y} = \frac{2y}{\sqrt{1-(x^2+y^2)^2}} - \frac{1}{y} + 2^{\frac{y}{x}} \ln 2 \frac{1}{x} \cdot 1$$

$$\frac{dy}{dx} = \frac{\partial f/\partial x}{\partial f/\partial y}$$ - after that we put values and get answer